andres@andres-X441NA: ~/Documentos/betek

```
andres@andres-X441NA
--------------------
OS: Linux Mint 22.3 x86_64
Host: X441NA 1.0
Kernel: 6.17.0-19-generic
Uptime: 1 hour, 11 mins
Packages: 2200 (dpkg), 16 (flatpak)
Shell: bash 5.2.21
Resolution: 1366x768
DE: Cinnamon 6.6.7
WM: Mutter (Muffin)
WM Theme: Orchis-Dark (Mint-Y)
Theme: Orchis-Dark [GTK2/3]
Icons: Papirus-Dark [GTK2/3]
Terminal: gnome-terminal
CPU: Intel Celeron N3350 (2) @ 2.400
GPU: Intel HD Graphics 500
Memory: 2057MiB / 3778MiB
Public IP: 181.78.65.238

andres@andres-X441NA:~/Documentos/betek$ docker pull nginx:latest
latest: Pulling from library/nginx
669e0ab8e7fa: Download complete
96a6cfe061e0: Download complete
510ddf6557d6: Pull complete
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
5e815e07e569: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
andres@andres-X441NA:~/Documentos/betek$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
CPU: Intel Celeron N3350 (2) @ 2.400
GPU: Intel HD Graphics 500
Memory: 2057MiB / 3778MiB
Public IP: 181.78.65.238

andres@andres-X441NA:~/Documentos/betek$ docker pull nginx:latest
latest: Pulling from library/nginx
669e0ab8e7fa: Download complete
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ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
5e815e07e569: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
andres@andres-X441NA:~/Documentos/betek$ docker images
                                                                                    1 Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   452a468a4bf9       25.9kB         9.49kB    U
nginx:latest         7150b3a39203        240MB         65.8MB
andres@andres-X441NA:~/Documentos/betek$ mkdir docker-lab
andres@andres-X441NA:~/Documentos/betek$ cd docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ <<EOF > Dockerfile
> FROM alpine:3.20
WORKDIR /app
RUN echo "Hola desde Docker" > /app/mensaje.txt
CMD ["cat", "/app/mensaje.txt"]
> EOF
andres@andres-X441NA:~/Documentos/betek/docker-lab$ ls
Dockerfile
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ <<EOF > Dockerfile
> FROM alpine:3.20
WORKDIR /app
RUN echo "Hola desde Docker" > /app/mensaje.txt
CMD ["cat", "/app/mensaje.txt"]
> EOF
andres@andres-X441NA:~/Documentos/betek/docker-lab$ ls
Dockerfile
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker build -t miapp:v1 .
[+] Building 3.0s (1/1) FINISHED                                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                                        1.4s
 => => transferring dockerfile: 31B                                                                                         0.3s
ERROR: failed to build: failed to solve: the Dockerfile cannot be empty
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker build -t miapp:v1 .
[+] Building 21.9s (7/7) FINISHED                                                                                 docker:default
 => [internal] load build definition from Dockerfile                                                                        0.2s
 => => transferring dockerfile: 147B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                              3.0s
 => [internal] load .dockerignore                                                                                           0.6s
 => => transferring context: 2B                                                                                             0.0s
 => [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805         3.1s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805         0.4s
 => => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB                              0.9s
 => => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd                                   0.7s
 => [2/3] WORKDIR /app                                                                                                      1.7s
 => [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt                                                                   3.7s
 => exporting to image                                                                                                      6.0s
 => => exporting layers                                                                                                     3.1s
 => => exporting manifest sha256:928fd338449bcec80b250b7c626260e324bde5cd91aab18b8957a8c9f194a7c7                           0.3s
 => => exporting config sha256:2c59dbecaf106ed5ee9b2b1d41a096de913d3b123a656caa89d5c08c7ac81ac9                             0.3s
 => => exporting attestation manifest sha256:d0577ccd8ba7c5ddfab26d86c4b3263d2848098337286e905fb06a167b28424f               0.7s
 => => exporting manifest list sha256:8eee7f34592e4c822c33b6e68c380bea2aeac2ee11f8aecbcf628ddfb6af1cf3                      0.3s
 => => naming to docker.io/library/miapp:v1                                                                                 0.1s
 => => unpacking to docker.io/library/miapp:v1                                                                              0.7s
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker run --rm miapp:v1
Hola desde Docker
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker tag miapp:v1 miapp:latest
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest         8eee7f34592e       12.1MB         3.63MB
miapp:v1             8eee7f34592e       12.1MB         3.63MB
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker image inspect miapp:v1
[
    {
        "Id": "sha256:8eee7f34592e4c822c33b6e68c380bea2aeac2ee11f8aecbcf628ddfb6af1cf3",
        "RepoTags": [
            "miapp:latest",
            "miapp:v1"
        ],
        "RepoDigests": [
            "miapp@sha256:8eee7f34592e4c822c33b6e68c380bea2aeac2ee11f8aecbcf628ddfb6af1cf3"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-03-26T19:18:58.485640624-05:00",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
            ],
            "Cmd": [
                "cat",
                "/app/mensaje.txt"
            ],
            "WorkingDir": "/app",
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 3631067,
        "RootFS": {
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
            "WorkingDir": "/app",
            "ArgsEscaped": true
        },
        "Architecture": "amd64",
        "Os": "linux",
        "Size": 3631067,
        "RootFS": {
            "Type": "layers",
            "Layers": [
                "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
                "sha256:decaea3fec153fb18593f3aa3bcd3bfd55780c4287463cf362f5af3e3e6c2dc5",
                "sha256:28ef51ce4825ddf86198547e3b9fdd445ce13b0b7870a1edff60a7931a2d7a02"
            ]
        },
        "Metadata": {
            "LastTagTime": "2026-03-27T00:22:04.922331245Z"
        },
        "Descriptor": {
            "mediaType": "application/vnd.oci.image.index.v1+json",
            "digest": "sha256:8eee7f34592e4c822c33b6e68c380bea2aeac2ee11f8aecbcf628ddfb6af1cf3",
            "size": 855
        },
        "Identity": {
            "Build": [
                {
                    "Ref": "81jvpntc220j6gt8omx9ks58o",
                    "CreatedAt": "2026-03-26T19:19:04.886720636-05:00"
                }
            ]
        }
    }
]
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker image inspect miapp:v1 --format '{{.Id}}'
sha256:8eee7f34592e4c822c33b6e68c380bea2aeac2ee11f8aecbcf628ddfb6af1cf3
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker save -o miapp.tar miapp:v1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ ls -lh miapp.tar
-rw------- 1 andres andres 3,5M mar 26 19:25 miapp.tar
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker rmi miapp:v1
Untagged: miapp:v1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker load -i miapp.tar
Loaded image: miapp:v1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest         8eee7f34592e       12.1MB         3.63MB
miapp:v1             8eee7f34592e       12.1MB         3.63MB
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker rmi miapp:latest
Untagged: miapp:latest
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1             8eee7f34592e       12.1MB         3.63MB
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker image prune -f
Total reclaimed space: 0B
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker run -d --name web1 nginx:latest
14069f517514e0d5fe23412596f47fdb14031349461c082e96430cfc9647cf70
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS     NAMES
14069f517514   nginx:latest   "/docker-entrypoint.…"   14 seconds ago   Up 12 seconds   80/tcp    web1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker run -d --name web2 -p 8080:80 nginx:latest
6176c20e862823cb5929a1c9cbe8ce4532fee05d9b3ff9027f224f0e0bb4b9cb
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps
curl http://localhost:8080
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAME
S
6176c20e8628   nginx:latest   "/docker-entrypoint.…"   15 seconds ago   Up 12 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
14069f517514   nginx:latest   "/docker-entrypoint.…"   47 seconds ago   Up 45 seconds   80/tcp                                      web1
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED              STATUS                     PORTS
          NAMES
6176c20e8628   nginx:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute          0.0.0.0:8080->80/tcp, [::]:8080-
>80/tcp   web2
14069f517514   nginx:latest   "/docker-entrypoint.…"   2 minutes ago        Up 2 minutes               80/tcp
          web1
9fffc1831816   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          kind_sutherland
d8101ea3efd3   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          nice_williams
7e8e36224059   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          charming_boyd
905ab63a95e3   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          tender_hawking
1ab48e2a4526   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          naughty_lederberg
5d7a24c21e48   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          gifted_neumann
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
          web1
9fffc1831816   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          kind_sutherland
d8101ea3efd3   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          nice_williams
7e8e36224059   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          charming_boyd
905ab63a95e3   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          tender_hawking
1ab48e2a4526   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          naughty_lederberg
5d7a24c21e48   hello-world    "/hello"                 46 hours ago         Exited (0) 46 hours ago
          gifted_neumann
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 00:32:24 [notice] 1#1: using the "epoll" event method
2026/03/27 00:32:24 [notice] 1#1: nginx/1.29.7
2026/03/27 00:32:24 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/27 00:32:24 [notice] 1#1: OS: Linux 6.17.0-19-generic
2026/03/27 00:32:24 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/03/27 00:32:24 [notice] 1#1: start worker processes
2026/03/27 00:32:24 [notice] 1#1: start worker process 29
2026/03/27 00:32:24 [notice] 1#1: start worker process 30
172.17.0.1 - - [27/Mar/2026:00:32:37 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.5.0" "-"
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker inspect web2 --format '{{.State.Status}}'
running
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/27 00:32:24 [notice] 1#1: using the "epoll" event method
2026/03/27 00:32:24 [notice] 1#1: nginx/1.29.7
2026/03/27 00:32:24 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/27 00:32:24 [notice] 1#1: OS: Linux 6.17.0-19-generic
2026/03/27 00:32:24 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/03/27 00:32:24 [notice] 1#1: start worker processes
2026/03/27 00:32:24 [notice] 1#1: start worker process 29
2026/03/27 00:32:24 [notice] 1#1: start worker process 30
172.17.0.1 - - [27/Mar/2026:00:32:37 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.5.0" "-"
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker inspect web2 --format '{{.State.Status}}'
running
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps -a | grep alpine1
4a7c1500cf7c   alpine:3.20    "sh"                     39 seconds ago   Exited (0) 20 seconds ago
        alpine1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker inspect web1 --format '{{.State.Paused}}'
false
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
      tender_hawking
1ab48e2a4526   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      naughty_lederberg
5d7a24c21e48   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      gifted_neumann
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker rm -f web1
web1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                     PORTS
      NAMES
6176c20e8628   nginx:latest   "/docker-entrypoint.…"   11 minutes ago   Up 11 minutes              0.0.0.0:8080->80/tcp, [::]:8080->80/
tcp   web2
9fffc1831816   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      kind_sutherland
d8101ea3efd3   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      nice_williams
7e8e36224059   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      charming_boyd
905ab63a95e3   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      tender_hawking
1ab48e2a4526   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      naughty_lederberg
5d7a24c21e48   hello-world    "/hello"                 46 hours ago     Exited (0) 46 hours ago
      gifted_neumann
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker rm -f web2
web2
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker ps -a
CONTAINER ID   IMAGE          COMMAND    CREATED        STATUS                     PORTS     NAMES
9fffc1831816   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            kind_sutherland
d8101ea3efd3   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            nice_williams
7e8e36224059   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            charming_boyd
905ab63a95e3   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            tender_hawking
1ab48e2a4526   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            naughty_lederberg
5d7a24c21e48   hello-world    "/hello"   46 hours ago   Exited (0) 46 hours ago            gifted_neumann
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ echo "<h1>Hola volumen</h1>" > index.html
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
andres@andres-X441NA:~/Documentos/betek/docker-lab$ curl http://localhost:8081
<h1>Hola volumen</h1>
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

SESIÓN 1 - CONTENEDORES | localhost:8081

localhost:8081

Hola volumen

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
andres@andres-X441NA:~/Documentos/betek/docker-lab$ echo "<h1>Hola volumen</h1>" > index.html
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
andres@andres-X441NA:~/Documentos/betek/docker-lab$ curl http://localhost:8081
<h1>Hola volumen</h1>
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker rm -f webvol
webvol
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker volume rm datos_nginx
datos_nginx
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker volume ls
DRIVER    VOLUME NAME
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
c637633a3c4d   bridge    bridge    local
8c9ab7345b51   host      host      local
7b30d48a3acd   none      null      local
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker network create mi_red
d308b3d0799e29c987fabd15f8f89ab56ba23678d3c8de7f537e44b138bb0ba3
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
c637633a3c4d   bridge    bridge    local
8c9ab7345b51   host      host      local
d308b3d0799e   mi_red    bridge    local
7b30d48a3acd   none      null      local
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker network inspect mi_red
[
    {
        "Name": "mi_red",
        "Id": "d308b3d0799e29c987fabd15f8f89ab56ba23678d3c8de7f537e44b138bb0ba3",
        "Created": "2026-03-26T19:57:19.84136591-05:00",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.18.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
afc9a3b64c4a70c07d2421aefd69e68f5c62f38d25d2bee0152d5c7f28309f27
5884d3219281a7655d038ba2f259d261c59d19c0fab393ec26f1da82ee2fce67
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.211 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.212 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.211/0.211/0.212 ms
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker network disconnect mi_red caja2
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
Global Options:
      --config string      Location of client config files (default "/home/andres/.docker")
  -c, --context string     Name of the context to use to connect to the daemon (overrides DOCKER_HOST env var and default
                           context set with "docker context use")
  -D, --debug              Enable debug mode
  -H, --host string        Daemon socket to connect to
  -l, --log-level string   Set the logging level ("debug", "info", "warn", "error", "fatal") (default "info")
      --tls                Use TLS; implied by --tlsverify
      --tlscacert string   Trust certs signed only by this CA (default "/home/andres/.docker/ca.pem")
      --tlscert string     Path to TLS certificate file (default "/home/andres/.docker/cert.pem")
      --tlskey string      Path to TLS key file (default "/home/andres/.docker/key.pem")
      --tlsverify          Use TLS and verify the remote
  -v, --version            Print version information and quit

Run 'docker COMMAND --help' for more information on a command.

For more help on how to use Docker, head to https://docs.docker.com/go/guides/
andres@andres-X441NA:~/Documentos/betek/docker-lab$ cat compose.yml
services:
    web:
        image: nginx:latest
        ports:
            - "8082:80"
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker-compose up -d
No se ha encontrado la orden «docker-compose», pero se puede instalar con:
sudo apt install docker-compose
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose version
Docker Compose version v5.1.1
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose up -d
[+] up 2/2
 ✔ Network docker-lab_default Created                                                                    0.3s
 ✔ Container docker-lab-web-1 Started                                                                    3.1s
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose ps curl http://localhost:8082
no such service: curl
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose ps curl http://localhost:8082
no such service: curl
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose ps
curl http://localhost:8082
NAME                IMAGE          COMMAND                  SERVICE   CREATED              STATUS               PORTS
docker-lab-web-1    nginx:latest   "/docker-entrypoint.…"   web       About a minute ago   Up About a minute    0.0.0.0:8082->80/tcp, [:
:]:8082->80/tcp
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/03/27 01:23:51 [notice] 1#1: using the "epoll" event method
web-1  | 2026/03/27 01:23:51 [notice] 1#1: nginx/1.29.7
web-1  | 2026/03/27 01:23:51 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/03/27 01:23:51 [notice] 1#1: OS: Linux 6.17.0-19-generic
web-1  | 2026/03/27 01:23:51 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
web-1  | 2026/03/27 01:23:51 [notice] 1#1: start worker processes
web-1  | 2026/03/27 01:23:51 [notice] 1#1: start worker process 29
web-1  | 2026/03/27 01:23:51 [notice] 1#1: start worker process 30
web-1  | 172.19.0.1 - - [27/Mar/2026:01:25:18 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.5.0" "-"
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose down
[+] down 2/2
 ✔ Container docker-lab-web-1 Removed                                                                    2.1s
 ✔ Network docker-lab_default Removed                                                                    0.4s
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab

```
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose up -d --scale web=3
[+] up 3/4
 ✔ Network docker-lab_default Created                                                                                    0.3s
 ✔ Container docker-lab-web-1 Created                                                                                    3.9s
 ✔ Container docker-lab-web-2 Started                                                                                    5.1s
 ⠿ Container docker-lab-web-3 Starting                                                                                   7.6s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-
lab-web-3 (474ca0b9a8161d71e4895e4d4ab897f4350c861b89aaac67160b315e1a869bf8): Bind for 0.0.0.0:8082 failed: port is already allocated
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose ps
NAME                IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-web-2    nginx:latest   "/docker-entrypoint.…"   web       39 seconds ago   Up 35 seconds   0.0.0.0:8082->80/tcp, [::]:8082-
>80/tcp
andres@andres-X441NA:~/Documentos/betek/docker-lab$
```

andres@andres-X441NA: ~/Documentos/betek/docker-lab/dockerfile-test

```
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose up -d --scale web=3
[+] up 3/4
 ✔ Network docker-lab_default Created                                                                                    0.3s
 ✔ Container docker-lab-web-1 Created                                                                                    3.9s
 ✔ Container docker-lab-web-2 Started                                                                                    5.1s
 ⠿ Container docker-lab-web-3 Starting                                                                                   7.6s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-
lab-web-3 (474ca0b9a8161d71e4895e4d4ab897f4350c861b89aaac67160b315e1a869bf8): Bind for 0.0.0.0:8082 failed: port is already allocated
andres@andres-X441NA:~/Documentos/betek/docker-lab$ docker compose ps
NAME                IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-web-2    nginx:latest   "/docker-entrypoint.…"   web       39 seconds ago   Up 35 seconds   0.0.0.0:8082->80/tcp, [::]:8082-
>80/tcp
andres@andres-X441NA:~/Documentos/betek/docker-lab$ mkdir dockerfile-test
andres@andres-X441NA:~/Documentos/betek/docker-lab$ cd dockerfile-test
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ touch archivo.txt
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ nano archivo.txt
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ cat archivo.txt
echo "contenido local"
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ touch Dockerfile
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ nano Dockerfile
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ cat Dockerfile
FROM alpine:3.20
LABEL autor="prueba"
ARG VERSION=1.0
ENV APP_ENV=dev
WORKDIR /app
COPY archivo.txt /app/archivo.txt
RUN echo "version=$VERSION" > /app/version.txt
ADD archivo.txt /app/archivo2.txt
VOLUME ["/data"]
CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat
/app/version.txt"]
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab/dockerfile-test
 => [internal] load build definition from Dockerfile                                                                    0.2s
 => => transferring dockerfile: 344B                                                                                    0.0s
Dockerfile:11
--------------------
   9 |     VOLUME ["/data"]
  10 |     CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat
  11 | >>> /app/version.txt"]
  12 |
--------------------
ERROR: failed to build: failed to solve: dockerfile parse error on line 11: unknown instruction: /app/version.txt"]
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ nano Dockerfile
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 18.4s (10/10) FINISHED                                                                          docker:default
 => [internal] load build definition from Dockerfile                                                                    0.2s
 => => transferring dockerfile: 344B                                                                                    0.0s
 => [internal] load metadata for docker.io/library/alpine:3.20                                                          0.5s
 => [internal] load .dockerignore                                                                                       0.9s
 => => transferring context: 2B                                                                                         0.0s
 => [1/5] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805     0.9s
 => => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805     0.8s
 => [internal] load build context                                                                                       1.2s
 => => transferring context: 75B                                                                                        0.0s
 => CACHED [2/5] WORKDIR /app                                                                                           0.0s
 => [3/5] COPY archivo.txt /app/archivo.txt                                                                             1.4s
 => [4/5] RUN echo "version=2.0" > /app/version.txt                                                                     2.9s
 => [5/5] ADD archivo.txt /app/archivo2.txt                                                                             1.7s
 => exporting to image                                                                                                  7.0s
 => => exporting layers                                                                                                 4.0s
 => => exporting manifest sha256:7f9b12dac1df12338b9ffc336a57555e0032a975a2c79682141fccb363d7b738                       0.3s
 => => exporting config sha256:c93165131d5510bd87443ef94d5278351457bf5ca36feef53a8427ea1a2943e6                         0.3s
 => => exporting attestation manifest sha256:35dcd7a8e5fa2818e31936012672ba35bb782d4ed207c3dad1332bcb596854fd           0.6s
 => => exporting manifest list sha256:d388a36b191f1ebb81f07e4ac3dda50109fae35e3934e6f315bae78abac87b6b                  0.3s
 => => naming to docker.io/library/dockerfile-demo:latest                                                               0.0s
 => => unpacking to docker.io/library/dockerfile-demo:latest                                                            1.0s
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$
```

```
andres@andres-X441NA: ~/Documentos/betek/docker-lab/dockerfile-test
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ docker run --rm dockerfile-demo
APP_ENV=dev
echo "contenido local" > archivo.txt
version=2.0
andres@andres-X441NA:~/Documentos/betek/docker-lab/dockerfile-test$ docker image inspect dockerfile-demo
[
    {
        "Id": "sha256:d388a36b191f1ebb81f07e4ac3dda50109fae35e3934e6f315bae78abac87b6b",
        "RepoTags": [
            "dockerfile-demo:latest"
        ],
        "RepoDigests": [
            "dockerfile-demo@sha256:d388a36b191f1ebb81f07e4ac3dda50109fae35e3934e6f315bae78abac87b6b"
        ],
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-03-26T20:38:10.162589353-05:00",
        "Config": {
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "APP_ENV=dev"
            ],
            "Cmd": [
                "sh",
                "-c",
                "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
            ],
            "Volumes": {
                "/data": {}
            },
            "WorkingDir": "/app",
            "Labels": {
                "autor": "prueba"
            },
            "ArgsEscaped": true
        },
```